

Chapter 4

Presentation

The many ways in which raw or derived data can be represented were discussed in the previous chapter. We now examine how these representations can best be **presented** – i.e., laid out on a display – for viewing and interaction by a user. While presentation (Fig. 4.1) is now the focus of our attention, its inevitable links with representation and interaction, as well as the characteristics of the user, will need to be taken into account.

When designing a presentation of represented data – in other words, its spatial and temporal layout – there are three principal resources available for beneficial exploitation by an interaction designer. One is the available display **space** which can range from that of a mobile to the vast area of a wall display. Another is the **time** available for the performance of the task that the presentation is designed to

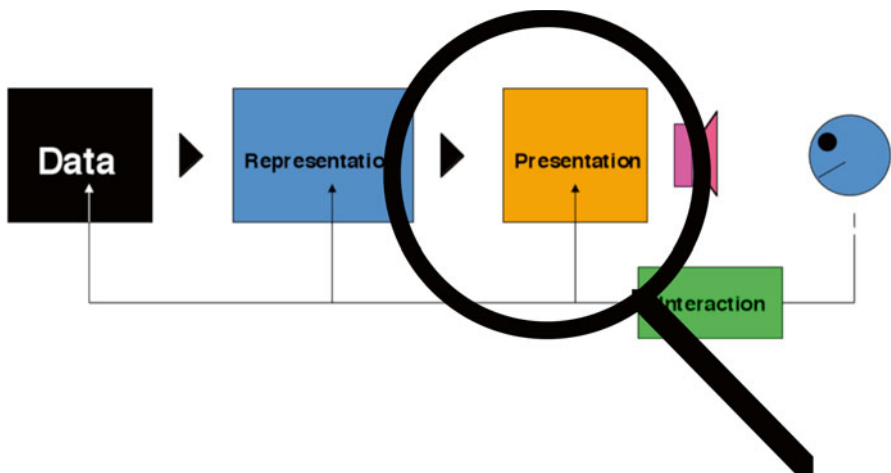


Fig. 4.1 The focus of this chapter is on presentation, but links to representation and interaction are important

support. A third comprises the characteristics of the human **user**. This last resource is fundamentally fixed by the very nature of the human visual processing system. Although its effectiveness can be modified, for example by training, it cannot fundamentally be changed by the designer and will therefore be discussed first. And because most data representations are visual we focus on the fascinating – and still not completely understood – properties of the human visual system.

4.1 Human Vision

Put very simply, a user *looks* at something and tries to *make sense* of what has been seen, processes broadly called *perception* and *cognition*.

4.1.1 Retinal Image

The scene in front of a user – the view out of a window or a mixture of text and images on a display screen – is projected, via the eye's corneal lens, onto the retina of each eye. While the appearance (Fig. 4.2) of a retina conveys little if anything of its function, more is identified by the three regions sketched in Fig. 4.3. In the centre is the foveal region of the retina within which part of an image can be seen in considerable detail and with good colour rendition. Around this foveal region, as one moves via the parafovea to the periphery, visual acuity – the ability to distinguish detail – falls off dramatically.

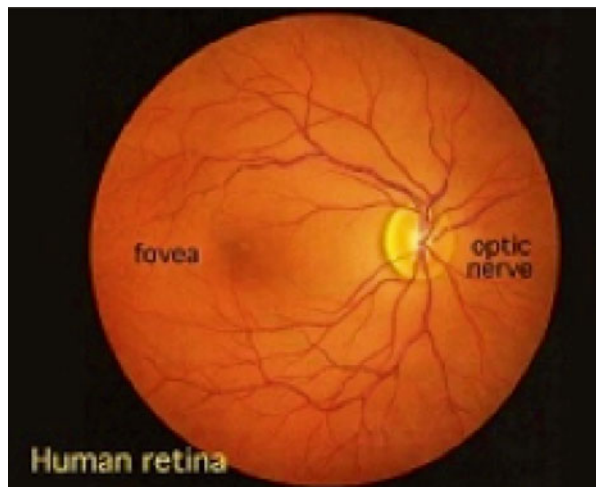
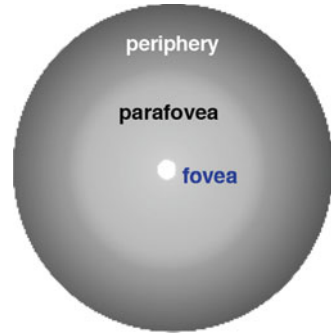


Fig. 4.2 The human retina.
Fovea is within the *dark*
region left of centre

Fig. 4.3 Functional regions of the retina (radial distance not to scale)



4.1.2 Fixations and Saccades

It is unfortunate, therefore, that the foveal region corresponds to a viewing angle of only about 2° (Fig. 4.4), an angle subtended at the eye by a fingernail held at arm's length (try it!). This very restricted detailed view of a scene is the reason why the eye must normally and quite rapidly be moved around so that different parts of the scene fall, in turn, on the fovea. A first 10-s view of a painting by Dali, for example (Fig. 4.5), resulted in a person's gaze (i.e., that part of a scene currently focused by the muscles of the eye on to the fovea) moving quite rapidly around the painting: each location (fixation) lasts around 200 ms, and the fixations are connected by very fast (e.g., 20 ms) saccades.

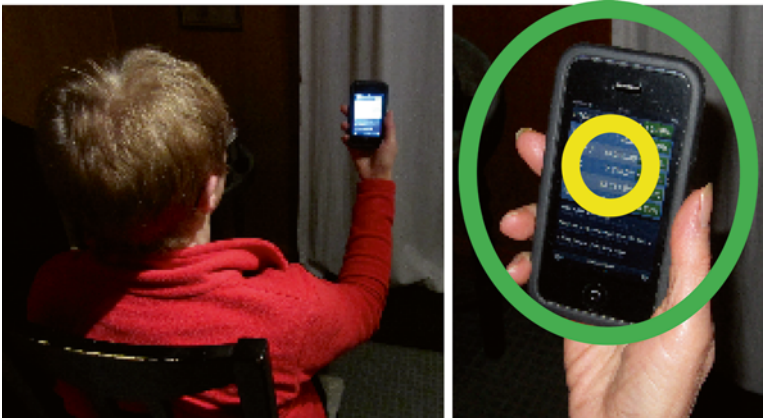


Fig. 4.4 Illustration of the extent of foveal (yellow) and parafoveal (green) vision

But precisely what causes gaze to be directed to a new location in a scene, whether that scene is a painting or a report containing diagrams? As gaze moves over a scene in a sequence of fixations and saccades (a procedure we shall simply call 'looking') the user may only be *glancing* at part of the scene rather than paying attention to it. After all, if fixations occur at the rate of about 5 per second, a 20 s



Fig. 4.5 Record of the eye-gaze locations (fixations **F**, lasting on average 200 mS) during a 10-s viewing of a Dali painting. Very fast transitions (saccades, approx 20 mS) between fixations are shown yellow

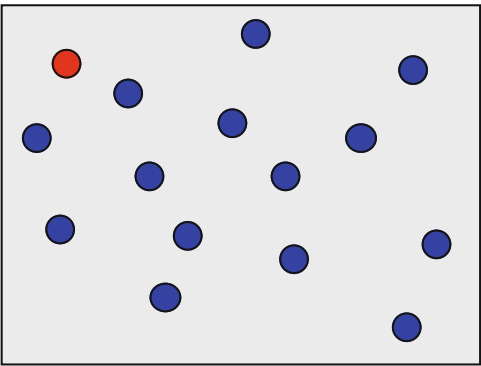
viewing of a painting involves around 100 fixations, and it is likely that careful attention – **overt attention** – is only being paid to a few of those locations. By contrast, simultaneous **covert attention**, sensitive to items within the parafoveal and peripheral regions, helps the brain to decide where gaze might next be usefully directed (e.g., to an unidentified person silently approaching).

4.1.3 Preattentive Processing

The last feature of the human visual processing system of which we need to be aware for the moment is a surprising and remarkable one, and concerns what happens within the first 200 ms (approx) of viewing a scene. It is best illustrated by three experiments. In the first (Treisman 1985, 1991; Treisman and Gormican 1988) the image shown in Fig. 4.6 is presented for only 50 ms on an otherwise blank screen. During that very brief glance a user will notice, and afterwards report, the presence of the single red dot among the many blue ‘distractors’. There is no time for a user to ‘pay attention’ in the normal sense of that phrase (after all, a fixation typically lasts for only 200 ms), which is why the phenomenon is referred to as ‘**preattentive processing**’. In the second experiment (Kundel and Nodine 1975) trained radiologists were presented with lung X-rays for a duration of only 200 ms: nevertheless, they were able to detect anomalies in 70 % of cases.¹ In the third

¹ Performance rose to 97 % under unlimited viewing conditions.

Fig. 4.6 A display lasting 50 mS, but nevertheless allowing a viewer to report the presence of a *red dot* among many *blue dots* (distractors)



experiment, Tse et al. (1998) found that users could successfully acquire the ‘gist’ of a film after viewing a sequence of twenty 100 ms frames specially selected from that film. Clearly, something unusual and potentially very beneficial is happening during this first short glimpse of a scene, and as interaction designers we must be ready to exploit that potential.

The properties of the human visual processing system we have discussed, and which are relevant to the presentation of represented data in space, are briefly summarised in Fig. 4.7. From what we know about preattentive processing and the typically rapid sequence of fixations and saccades involved in looking at a scene, it might be safe to assume that the ‘understanding’ of a scene takes on the qualitative form sketched in Fig. 4.8, starting with a step function during preattentive processing.²

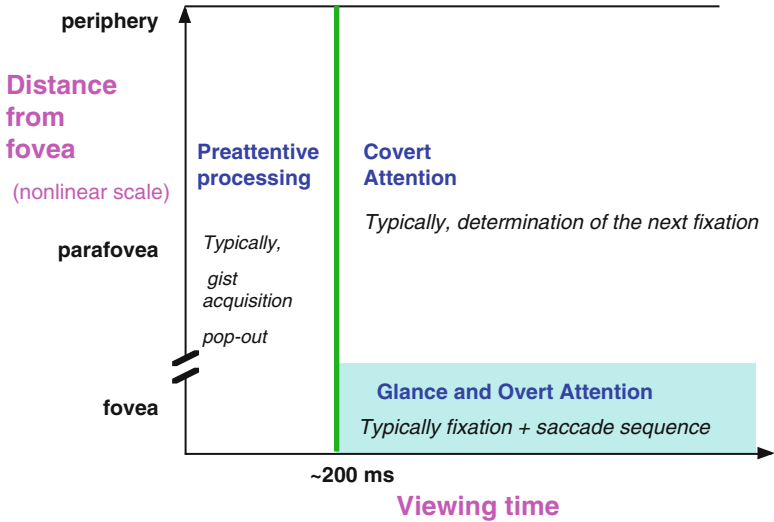
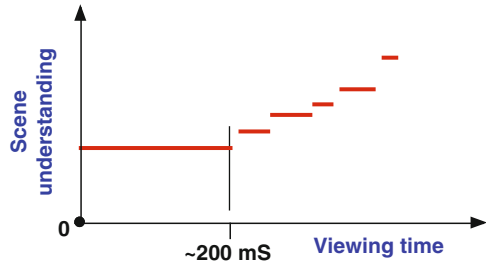


Fig. 4.7 A summary of properties of the human visual processing system particularly relevant to the stage of presentation

²The description of the human visual processing system given here is necessarily a gross simplification of current understanding (see, for example, Findlay and Gilchrist 2003), but is nevertheless adequate and useful for the interaction designer.

Fig. 4.8 Suggested qualitative variation of the variation of scene understanding with viewing time



4.1.4 Summary

We might usefully summarise the foregoing discussion as follows:

The narrow field of view associated with the detailed viewing and comprehension of part of an image requires that glance and sometimes overt attention be moved from one part of a scene to another, a movement achieved either by rapid eye movements or by interaction that brings a different image or part of an image into focus. At the same time, covert attention to those parts of a scene falling outside the foveal region helps to decide where gaze should next be focused. In the first viewing of a scene – up to about 200 ms – preattentive processing can capture many features of that entire scene.

In what follows next in Sect. 4.2 we shall see how limits to display area (*‘space’*) influence the design of presentation, though we shall additionally discover the need to consider the influence of representation and interaction. In Sect. 4.3 we then consider the influence of presentation *time*. Section 4.4 then investigates how both space and time can be exploited, in conjunction with the remarkable properties of the human visual system, to the benefit of the information visualization process. A very brief Sect. 4.5 concerns the representation of presentation.

4.2 Presentation in Space

Let’s suppose that I’m composing and editing a page of a report (Fig. 4.9), and that it appears full-size on a display of conventional (e.g., laptop) size. I can easily glance at the content by moving my gaze around the page in order to foveate on and, if appropriate, attend in detail to and interact with, an item of particular interest. And although I may recall the locations of various figures and other items (because I put them there) my parafoveal and peripheral vision helps me, through covert attention, to decide where next to move my gaze in order to pay